Partial Solutions To Exercises In Calculous 3

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Intro

Partial solution to exercises from 26A calc3 homeworks. Learned using Shiri's 25A course notes and Boris Tsirelson lectures from 2016.

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. . . . . (1) . . . . .

1. Suppose $A \in \hookleftarrow (\mathbb{R}^n, \mathbb{R}^m)$ and $B \in \hookleftarrow (\mathbb{R}^n, \mathbb{R}^m)$ are matrices. We'll prove HS norm and the operator norm a sub-multiplicative.

   Proof.
   - **Operator norm:** Let $x \in \mathbb{R}^n$. Then by the very definition of $||A||_{op}$ and $||B||_{op}$ we get that:

     $$||ABx|| \leq ||A|| \cdot ||Bx|| \leq ||A|| \cdot ||B|| \cdot ||x||$$

     Hence we get that:

     $$\forall x \in \mathbb{R}^n \colon \frac{||ABx||}{||x||} \leq ||A||_{op} \cdot ||B||_{op}$$

     Therefore, the supremum on all $x \in \mathbb{R}^n$, which is defined to be $||AB||_{op}$, is bounded by $||A||_{op} \cdot ||B||_{op}$ as intended.
   - **HS:** We'll alternatively write:

     $$||M||_{HS} = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{k} M_{ij}^2} = \sqrt{\sum_{i=1}^{n} ||Me_i||_2^2}$$

     We clearly get for every matrix M that:

     $$||M||_{HS}^2 = \sum_{i=1}^{n} ||Me_i||_2^2 \geq \sum_{i=1}^{n} (\left|\left|M^{-1}\right|\right|_{op})^{-1}$$

     Now:

     $$||AB||_{\text{HS}}^2 = \sum_{i=1}^{n} ||A(Be_i)||_2^2 \leq \sum_{i=1}^{n} ||A||_{\text{op}}^2 ||Be_i||_2^2 = ||A||_{\text{op}}^2 ||B||_{\text{HS}}^2$$

     It remains to prove that $||M||_{\text{op}} \leq ||M||_{\text{HS}}$ for each matrix M. The SVD decomposition of M yields orthonormal bases $u_1 \dots u_n$ and $v_1 \dots v_n$ s.t. $Mx = \sum_{i=1}^{r} \sigma_i v_i u_i^T \cdot x$ (where r is the rank). Now, assuming $r \neq 0$:

     $$||M||_{\text{HS}}^2 = \sum_{i=1}^{n} ||Me_i||^2 = \sum_{i=1}^{n} \left|\left| \sum_{i=1}^{r} \sigma_i v_i u_i^T \cdot e_i \right|\right|^2 = m \sum_{i=1}^{n} \sigma_i^2 \underbrace{\left|\left| v_i u_i^T \right|\right|}_{1} \geq r \cdot \max_i \sigma_i^2 = m\, ||M||_{\text{op}}^2 \geq ||M||_{\text{op}}$$

     Otherwise $r = 0$ and then $||M||_{\text{HS}} = ||M||_{\text{op}} = 0$ and we finished. Finally $||AB||_{\text{HS}}^2 \leq ||A||_{\text{op}}^2 ||B||_{\text{HS}}^2 \leq ||A||_{\text{HS}}^2 ||B||_{\text{HS}}^2$ ergo $||AB||_{\text{HS}} \leq ||A||_{\text{HS}} ||B||_{\text{HS}}$ as intended.

   ■

2. Let $A \in \hookleftarrow (\mathbb{R}^n, \mathbb{R}^m)$ a linear operator with $u_1 \dots u_n$ orthonormal basis for $\mathbb{R}^n$. We'll prove $||A||_{\text{HS}}^2 = \sum_{i=1}^{n} ||Mu_i||^2$.

   Proof. For $u_i = e_i$ it follows from the definition. Otherwise, there exists some unitary matrix U s.t. $Ue_i = u_i$. Now $||Mu_i|| = ||MUe_i|| = ||Me_i||$ (as composing by a unitary matrix does not change a vector's norm). This finishes the proof. ■

3. We'll show that the parallelogram law holds for every norm induced from an inner product structure.

   Proof. Consider $N(x) = \sqrt{\langle x \,|\, x \rangle}$. Then:

   $$\begin{aligned}
   N(x+y)^2 + N(x-y)^2 &= \langle x+y \,|\, x+y \rangle + \langle x-y \,|\, x-y \rangle \\
   &= \langle x \,|\, x+y \rangle + \langle y \,|\, x+y \rangle + \langle x \,|\, x-y \rangle - \langle y \,|\, x-y \rangle \\
   &= \langle x \,|\, x \rangle + \langle x \,|\, y \rangle + \langle y \,|\, x \rangle + \langle y \,|\, y \rangle \\
   &\quad + \langle x \,|\, x \rangle - \langle x \,|\, y \rangle - \langle y \,|\, x \rangle + \langle y \,|\, y \rangle \\
   &= 2 \langle x \,|\, x \rangle + 2 \langle y \,|\, y \rangle + \cancel{2 \langle x \,|\, y \rangle - 2 \langle y \,|\, x \rangle} \\
   &= 2N(x)^2 + 2N(y)^2
   \end{aligned}$$

   As intended. ■

4. Consider $D \subset M_{2\times 2}(\mathbb{R})$ be the set of diagonalizable matrices. Then:
   - **Closure:** consider some $A \in M_{2\times 2}(\mathbb{R})$. If A is diagonalizable, we finished. Otherwise, it's similar to Jordan $J_2(\lambda)$. Now the sequence of $J\varepsilon(\lambda) \coloneqq \left(\begin{smallmatrix}\lambda & 1\\ \varepsilon & \lambda\end{smallmatrix}\right)$ for $\varepsilon \to 0$ clearly converge to $J_2(\lambda)$. Moreover, $J_\varepsilon(\lambda)$ are diagonalizable matrices as their characteristic polynomial is $(\lambda - x)^2 - 1 \cdot \varepsilon = \lambda^2 - 2\lambda x + x^2 + \varepsilon = 0$ so:
   $$x = \frac{2\lambda \pm \overbrace{\sqrt{4 - 4\lambda^2\varepsilon}}^{>0}}{2}$$
   We got two distinct eigenvalues for $J_\varepsilon(\lambda)$ hence it's diagonalizable. Now when $A = UJ_2(\lambda)U^{-1}$ we know that $A_\varepsilon \coloneqq UJ_\varepsilon(\lambda)U^{-1}$ for $\varepsilon = \frac{1}{n}$ is a sequence of diagonalizable matrices. Moreover, it approaches A since:
   $$\begin{aligned}||A - A_\varepsilon||_{\text{op}} &= \left|\left|UJ_2(\lambda)U^{-1} - UJ_\varepsilon(\lambda)U^{-1}\right|\right|_{\text{op}} = \left|\left|U(J_2(\lambda) - J_\varepsilon(\lambda))U^{-1}\right|\right|_{\text{op}}\\ &\le \underbrace{||U||_{\text{op}}\left|\left|U^{-1}\right|\right|_{\text{op}}}_{1} ||J_2(\lambda) - J_\varepsilon(\lambda)||_{\text{op}}\\ &= \left|\left|\begin{pmatrix}0 & 0\\ \varepsilon & 0\end{pmatrix}\right|\right|_{\text{op}} \le \left|\left|\begin{pmatrix}0 & 0\\ \varepsilon & 0\end{pmatrix}\right|\right|_{\text{HS}} = \varepsilon \to 0\end{aligned}$$
   We got that A is a limit point of a sequence in D. Therefore $A \in \operatorname{Cl} D$. Finally $\operatorname{Cl} D = M_{2\times 2}(\mathbb{R})$.
   - **Interior:** the complement of D is dense too. The proof is similar, except we're using $A_\varepsilon = U\left(\begin{smallmatrix}\lambda_1 & \varepsilon\\ 0 & \lambda_2\end{smallmatrix}\right)U^{-1}$ instead (when $A \sim \operatorname{diag}(\lambda_1, \lambda_2)$). Since $\operatorname{Cl} X \setminus D = X$ then $\operatorname{Int} D = X \setminus \operatorname{Cl}(X \setminus D) = X \setminus X = \varnothing$.
   - **Boundary:** clearly $\partial D = \operatorname{Cl} D \setminus \operatorname{Int} D = M_{2\times 2}(\mathbb{R})$.
5. Directly from the definition.

. . . . . (2) . . . . .

1. Suppose for two norms N_1, N_2 the limit $\lim_{x\to 0} \frac{N_2(x)}{N_1(x)} = L$ exists. We'll prove $N_2 = L \cdot N_1$.

   Proof. Let $x \in \mathbb{R}^d$. If $x = 0$, we finished. Otherwise, for each $\hat{\varepsilon}$ there is some δ s.t. $-\hat{\varepsilon} + L \le \frac{N_2(x)}{N_1(x)} \le L + \hat{\varepsilon}$. Let $\varepsilon > 0$ and consider $\hat{\varepsilon} = \frac{\varepsilon}{N_1(x)}$. We deduce from above that $|N_2(x) - L \cdot N_1(x)| \le \varepsilon$ for all $\varepsilon \in \mathbb{R}$. Hence $N_2(x) - L \cdot N_1(x) = 0$ for each $x \in \mathbb{R}^d$, and we got $N_1 = L \cdot N_2$. ∎

2. Let $X \subset \mathbb{R}^n$ be a path-connected set, and x_0 an accumulation point of X. Let $f\colon X \to \mathbb{R}$. We'll prove that $\lim_{x\to x_0} f(x) = y$ iff for any $\gamma\colon (0,1) \to X \setminus \{x_0\}$ continues with $\lim_{t\to 0^+} \gamma(t) = x_0$, we get that $\lim_{t\to 0^+} f(\gamma(t)) = y$.

   Proof. $\implies$ Suppose that $\lim_{x\to x_0} f(x) = y$. Let γ be such path. Let $0 < t_n \to 0$ be a sequence in $\mathbb{R}$ (WOLOG bounded by 1). Now, $x_n \coloneqq \gamma(t_n)$ is a sequence if $X \setminus \{x_0\}$, and because $\lim_{t\to 0^+} \gamma(t) = x_0$ we know that $x_0 \ne x_n \to x_0$. Since $\lim_{x\to x_0} f(x) = y$ and $x_0 \ne x_n \to x_0$, we know that $f(x_n) \to y$ too. Now for each t_n approaching 0 from above we got that $f(\gamma(t_n)) = f(x_n) \to y$ and from Heine we get that $\lim_{t\to 0^+} f(\gamma(t)) = y$.

   $\impliedby$ Suppose that any such γ has the property $\lim_{t\to 0^+} f(\gamma(t)) = y$. Consider some $x_n \to x_0$ with $x_n \ne x_0$. As each $x_i, x_{i+1} \in X$ and X is path-connected there exists a continues path $\gamma_i\colon [0,1] \to X$ s.t. $\gamma_i(0) = x_i \wedge \gamma_i(1) = x_{i+1}$. Now consider the infinite sum $\gamma \coloneqq \sum_{i=1}^{\infty} \gamma_i$ defined as:
   $$\gamma \coloneqq \sum_{i=1}^{\infty} \gamma_i \coloneqq \lambda x \in (0,1).\begin{cases}\gamma_1(2x-1) & x \in \left[\frac{1}{2}, 1\right]\\ \gamma_2(4x-1) & x \in \left[\frac{1}{4}, \frac{1}{2}\right]\\ \vdots & \vdots\\ \gamma_n(2^n x - 1) & x \in \left[\frac{1}{2^n}, \frac{1}{2^{n-1}}\right]\end{cases}$$
   Note that this is well-defined as if $x = \frac{1}{2^n}$ then $\gamma_n(2^n x - 1) = \gamma_n(0) = \gamma_{n+1}(1) = \gamma_{n+1}(2^{n+1}x - 1)$. For this reason γ is also continues and for each $n \ne 0, 1$ we get that $\gamma(2^{-n}) = x_n$. Since γ is continues and we have partial limit of x_0, γ must converge to x_0 for $t \to 0^+$. Now $\lim_{t\to 0^+} f(\gamma(t)) = y$ therefore $f(\gamma(\frac{1}{2^n})) = f(x_n)$ must converge to y too, as intended.

   ∎

3. Let $E \subset X$. We'll prove that the indicator $\mathbb{1}_E$ is discontinues on $x \in X$ iff $x \in \partial E$.

Proof. If $\partial E = \varnothing$, we finished. Otherwise:

$\implies$ Suppose that $\mathbb{1}_E$ is discontinues at x. Then either the limit does not exists or

$\impliedby$ Suppose that $x \in \partial E = \operatorname{Cl} E \cap \operatorname{Cl}(X \setminus E)$. Then $x \in \operatorname{Cl} E$ therefore there exists a sequence $x_n \to x$ s.t. $x_n \in X$. In addition there is a sequence $y_n \to x$ s.t. $y_n \in X \setminus E$. By definition $f(y_n) = 0 \wedge f(x_n) = 1$ therefore from Heine the limit $\lim_{y \to x} f(y)$ does not exists.

■

4. (a) Consider the function $f(x,y) = \frac{x+y}{\log(x^2+y^2)}$. We'll prove when $(x,y) \to (0,0)$ then $f(x,y) \to 0$. From AMGM we know that $x^2 + y^2 \geq 2xy$. Since log monotonically increases, we get that $\log(x^2+y^2) \geq \log(2xy)$, and $\left|x^2\right| \leq |\log x|$ for small enough x ($x < 0.5$) and therefore:

$$|f(x,y)| = \left|\frac{x+y}{\log(x^2+y^2)}\right| \leq \left|\frac{x+y}{\log(2xy)}\right| = \left|\frac{x+y}{\log 2x + \log 2y}\right| \overset{!}{\leq} \left|\frac{x+y}{4x^2+4y^2}\right| \to 0$$

($\overset{!}{\leq}$ is true only once (x,y) reaches certain bound)

6. Let $\varnothing \neq A \subset \mathbb{R}^d$. We'll prove that if any continues $f \colon A \to \mathbb{R}$ is bounded, then $f \colon A \to \mathbb{R}$ attains maximum & minimum.

Proof. $\implies$ AFSOC A is not closed and bounded.

- If A if not bounded, then for $f(x) = x$ the identity, f is not bounded although its continues which is a contradiction.
- If A if not closed, then $\operatorname{Cl} A \setminus A \neq \varnothing$. Therefore there is some point $x_0 \in \operatorname{Cl} A \setminus A$, which implies the existence of a sequence $x_n \to x_0$ s.t. $x_n \in A$. WOLOG $||x_n||$ monotonically increases (otherwise we can always choose a monotonic subsequence of x_n instead, and for monotonically decreasing sequences the solution is similar). Now define f as follows:

$$f(x) = \begin{cases} \frac{1}{n-1} + \left(||x|| - \frac{1}{n}\right) & ||x|| \in [||x_{n-1}||, ||x_n||] \\ \inf A + ||x|| & ||x|| \in [\inf A, x_1] \\ ||x_0 + (x - v)|| & ||x|| \in [||x_0||, \sup ||A||] \end{cases}$$

(Where v is some $v \in \mathbb{R}^d$ s.t. $||v|| = \sup A$) WOLOG A is bounded otherwise we proved a contradiction. From the definition of linear interpolation, and because $||x_0|| = \sup ||x||_n$, it's clear that f is continues. Moreover, $f(x_n) = \frac{1}{n}$ and therefore f is not bounded.

Finally A is closed and bounded and from Heine-Borel it's also compact. Therefore $f(A)$ is compact too and we finished.

$\impliedby$ If each $f \colon A \to \mathbb{R}$ attains a maximum and minimum, it's in particular bounded.

■

7. Suppose $\varnothing \neq A, B \subset \mathbb{R}^d$ are closed and disjoint.

(a) We'll prove $x \mapsto d(x,A)$ is continues and $d(x,A) = 0$ iff $x \in A$.

Proof. If $x \in A$ clearly $d(x,A) = 0$. Otherwise since A is closed there is an basic open neighborhood of x s.t. $x \in U = B(x,\varepsilon)$ and $U \cap A = \varnothing$. Now clearly $d(x,A) \geq \varepsilon > 0$ and we finished.

We'll proceed to prove that $x \mapsto d(x,A)$ is continues. For each $\varepsilon > 0$ exists N s.t. $d(x_{n>N}, x) < \varepsilon$. Let $\varepsilon > 0$ and choose such N. Now:

$$\begin{aligned} d(x,A) \leq d(x,x_n) + d(x_n,A) \quad &\implies \quad d(x,A) - d(x_n,A) \leq \varepsilon \\ d(x_{n_k},A) \leq d(x_n,x) + d(x,A) \quad &\implies \quad d(x_n,A) - d(x,A) \leq \varepsilon \end{aligned}$$

Therefore $|d(x_n,A) - d(x,A)| \leq \varepsilon$ for $n > N$. By definition $d(x_n,A) \to d(x,A)$ as intended. ■

(b) Clearly exists such Urysohn function because $\mathbb{R}^d$ is completely normal. See topology exercises.

. . . . . (3) . . . . . .

1.

2. Let $\gamma, \eta \colon \mathbb{R} \to \mathbb{R}^n$ to differentiable functions. We'll prove that:

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle \gamma(t) \,|\, \eta(t) \rangle = \langle \gamma'(t) \,|\, \eta(t) \rangle + \langle \gamma(t) \,|\, \eta'(t) \rangle$$

Proof. Denote $h(t) = \gamma(t)^T \eta(t)$. Then:

$$\begin{aligned} h(t+s) - h(t) &= \langle \gamma(t+s) \,|\, \eta(t+s) \rangle - \langle \gamma(t) \,|\, \eta(t) \rangle \\ &= \langle \gamma(t+s) \,|\, \eta(t+s) \rangle - \langle \gamma(t+s) \,|\, \eta(t) \rangle - \langle \gamma(t) \,|\, \eta(t) \rangle + \langle \gamma(t+s) \,|\, \eta(t) \rangle \\ &= \langle \gamma(t+s) \,|\, \eta(t+s) - \eta(t) \rangle + \langle \gamma(t+s) - \gamma(s) \,|\, \eta(t) \rangle \end{aligned}$$

And by definition:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} h(t) &= \lim_{s \to 0} \frac{h(t+s) - h(t)}{s} = \lim_{s \to 0} \frac{\langle \gamma(t+s) \,|\, \eta(t+s) - \eta(t) \rangle + \langle \gamma(t+s) - \gamma(s) \,|\, \eta(t) \rangle}{s} \\ &= \left(\lim_{s \to 0} \gamma(t+s) \right)^T \left(\lim_{s \to 0} \frac{\eta(t+s) - \eta(t)}{s} \right) + \left(\lim_{s \to 0} \frac{\gamma(t+s) - \gamma(s)}{s} \right)^T \eta(t) \\ &= \gamma(t)^T \eta'(t) + \gamma'(t)^T \eta(t) = \langle \gamma(t) \,|\, \eta'(t) \rangle + \langle \gamma'(t) \,|\, \eta'(t) \rangle \end{aligned}$$

And if $||\gamma(t)||$ is constant $\sqrt{c}$, then $c = ||\gamma(t)||^2 = \langle \gamma(t) \,|\, \gamma(t) \rangle$ is constant too, which happens if and only if its derivative is 0, and by the theorem above:

$$0 = \langle \gamma \,|\, \gamma' \rangle + \langle \gamma' \,|\, \gamma \rangle = 2 \langle \gamma' \,|\, \gamma \rangle \iff \langle \gamma' \,|\, \gamma \rangle = 0 \iff \gamma' \perp \gamma$$

■

3. Let $f, g \colon \mathbb{R}^n \to \mathbb{R}$ by differentiable functions. We'll prove $f \cdot g$ is differentiable too, and find $\nabla(f \cdot g)$.

Proof. Denote $h = f \cdot g$. Then:

$$\begin{aligned} h(x+v) - h(x) &= f(x+v)g(x+v) - f(x)g(x) \\ &= f(x+v)g(x+v) - f(x+v)g(x) - f(x)g(x) + f(x+v)g(x) \\ &= f(x+v)\big(g(x+v) - g(x)\big) + g(x)\big(f(x+v) - f(x)\big) \end{aligned}$$

Consider the linear transformation $A_x \coloneqq f(x)\nabla(g) + g(x)\nabla(f)$. Now (no need for $|\cdot|$ since $|P| \to 0$ iff $p \to 0$):

$$\begin{aligned} \lim_{v \to 0} \frac{|h(x+v) - h(x) - Av|}{||v||} &= \lim_{v \to 0} \frac{f(x+v)\big(g(x+v) - g(x)\big) + g(x)\big(f(x+v) - f(x)\big) - Av}{||v||} \\ &= \lim_{v \to 0} f(x+v) \frac{g(x+v) - g(x)}{||v||} \end{aligned}$$

■

4.

5. Let $f \colon \mathbb{R}^n \to \mathbb{R}^m$ be a differentiable function. Suppose $W \subset \mathbb{R}^m$ is a linear space s.t. $\operatorname{Im} f \subset W$. We'll prove $\operatorname{Im}(D_f x) \subset W$.

Proof. Let $x \in \mathbb{R}^n$.Ad absurdum there exists $y \in \mathbb{R}^n$ s.t. $(D_f x)y \notin W$. Then for all $\varepsilon \in \mathbb{R}$ we know that $(D_f x)(\varepsilon \cdot y) \notin W$ from the linearity of $D_f x$. Now notice that:

$$0 = \lim_{x' \to x} \frac{||f(x') - f(x) - (D_f x)(x - x_0)||}{||x - x'||} = \lim_{y' \to 0} \frac{||f(x+y') - f(x) - (D_f x)(y')||}{||y||}$$

And in particular, for the subsequence $\varepsilon_n = \frac{1}{n}$ and $y_n = \varepsilon_n \cdot y$. Then:

$$0 = \lim_{n \to \infty} \frac{||\overbrace{f(x+y_n) - f(x)}^{\coloneqq p_n \in W} - \overbrace{(D_f x)(y_n)}^{\coloneqq q_n \notin W}||}{||y_n||} = ||y|| \cdot \lim_{n \to \infty} \frac{||p_n - q_n||}{\frac{1}{n}} = \lim_{n \to \infty} n\,||p_n - q_n|| = \lim_{n \to \infty} ||np_n - (D_f x)(y)||$$

Therefore $np_n \xrightarrow{n \to \infty} (D_f x)(y) \notin W$, although $np_n \in W$ (linearity) so $\lim_{n \to \infty} np_n \in W$ (W is a closed subset of $\mathbb{R}^n$, as a linear space), which forms a contradiction. ■

6. Suppose that $f\colon \mathbb{R}^n \to \mathbb{R}^n$ is differentiable at the origin and $f(\lambda x) = \lambda f(x)$ for all $\lambda \in \mathbb{R}$. We'll prove that f is linear.

   Proof. Clearly $f(0) = 0$ since $f(0) = f(0 \cdot 0) = 0f(0) = 0$. Denote $D_f \coloneqq D_f 0$. Now for all x, y we know that $\left|\left|\frac{f(0)-f(x+y)-D_f\cdot(x+y)}{x+y}\right|\right| \to 0$ as $x+y$ approaches 0, and considering $f(0) = 0$ and the linearity of D_f we get that $\left|\left|\frac{f(x+y)-((D_f)x+(D_f)y)}{x+y}\right|\right| \to 0$. In particular for all of the λ-multiplicities of x, y. It immediately follows from linearity of D_f and from the fact that $f(\lambda x) = \lambda f(x)$ that:

$$\begin{aligned} 0 = \lim_{x+y\to 0}\left|\left|\frac{f(x+y)-((D_f)x+(D_f)y)}{x+y}\right|\right| &\overset{!}{=} \lim_{\lambda\to 0}\left|\left|\frac{\lambda f((x+y))-\lambda\left((D_f)x+(D_f)y\right)}{\lambda\left(x+y\right)}\right|\right| \\ &= \left|\left|\frac{f((x+y))-((D_f)x+(D_f)y)}{(x+y)}\right|\right| \end{aligned}$$

   Where $\overset{!}{=}$ is true as we're taking a subsequence of $x + y \to 0$. It implies that $f(x + y) = (D_f)x + (D_f)y$ therefore f is linear. ∎

7. Consider $F(A) = A^2$ ($F\colon M_{n\times n}(\mathbb{R}) \hookleftarrow$). We'll show by definition that F is differentiable with $[D_f A](H) = AH + HA$.

   Proof. We need to prove that $\lim_{H\to 0} \frac{||f(A+H)-f(A)-[D_f A](H)||}{||H||} = 0$. By declaring $[D_f A](H) = AH + HA$ this is clear since:

$$\frac{\left|\left|(A+H)^2 - A^2 - AH - HA\right|\right|}{||A-H||} = \frac{\left|\left|H^2\right|\right|}{||A-H||}$$

   For all $A \neq 0$, A has a singular value greater than 0 and then $\lim_{H\to 0} ||A-H|| = ||A|| \neq 0$. In this case:

$$\lim_{H\to 0} \frac{||f(A+H)|| - f(A) - [D_f A](A)}{||H||} = \lim_{H\to 0} \frac{||H||}{||A-H||} = \frac{0}{||A||} = 0$$

   If $A = 0$ then we get the limit: [all norms are equal, so we'll use the operator norm]

$$\lim_{H\to 0} \frac{\left|\left|H^2\right|\right|}{||A-H||} = \lim_{H\to 0} \frac{||H||^2}{||H||} = \lim_{H\to 0} ||H|| = 0$$

   Either way the limit approaches 0 as intended. ∎

. . . . . (4) . . . . .

1.

2. Suppose $T \in C^1(\mathbb{R}^2)$ satisfies the Cauchy-Riemann equations. We'll call such T an *holomorphic function*.

   (a) We'll show that at any $p \in \mathbb{R}^2$ we got wither $D_T(p) > 0$ or $D_T(p) = 0$.

   Proof. We'll find the Jacobian of such a function. Denote $a = \frac{\partial T_1}{\partial x_1}(p)$. Then $\frac{\partial T_2}{\partial x_2}(p) = a$ by the Cauchy-Riemann equations. Similarly if we denote $b = \frac{\partial T_1}{\partial x_2} = b$ then we got $-b = \frac{\partial T_2}{\partial x_1}$. Then one would represent $D_T(p)$ by the standard base $\mathcal{E} = (x_1, x_2)$ and get the Jacobian:

$$[D_T(p)]_{\mathcal{E}} = \begin{pmatrix} \dfrac{\partial T_1}{\partial x_1} & \dfrac{\partial T_1}{\partial x_2} \\ \dfrac{\partial T_2}{\partial x_1} & \dfrac{\partial T_2}{\partial x_2} \end{pmatrix} = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

   Therefore $\det D_T(p) = \det[D_T(p)]_{\mathcal{E}} = a^2 - b(-b) = a^2 + b^2 \geq 0$. Now clearly $\det D_T(p) = 0$ iff $a = b = 0$ and in such case $D_T(p) = 0$. Otherwise $a^2 + b^2 > 0$. ∎

   (b) We'll show that the composition of two holomorphic functions is holomorphic too.

   Proof. We know that $D_(T \cdot S) = TD_S + SD_T$. It's clear that there exists $a, b, c, d \in \mathbb{R}$ s.t.:

$$[D_{T\cdot S}]_{\mathcal{E}} = T(p) \cdot \begin{pmatrix} a & b \\ -b & a \end{pmatrix} + S(p) \cdot \begin{pmatrix} c & d \\ -d & c \end{pmatrix}$$

   Now denote $\ell = T(p)a + S(p)c$ and $m = T(p)b + S(p)d$. Now clearly $[D_{T\cdot S}]_{\mathcal{E}} = \left(\begin{smallmatrix} c & d \\ -d & c \end{smallmatrix}\right)$... ∎

3. **Hadmard's Lemma:** Let $U \subset \mathbb{R}^n$ be a convex domain, and let $a \in U$ and $f \in C^1(U, \mathbb{R})$. Then there exists $g_1 \dots g_n \in C^1(U, \mathbb{R})$ s.t. $f(x) = f(a) + \sum_{i=1}^n (x_i - a) g_i(x)$. We'll prove that $g_i(a) = \frac{\partial f}{\partial x_i}$.

   Proof. hmm that's kinda trivial since:

$$D[f] = D\left[f(a) + \sum_{i=1}(x_i - a)g_i\right] = D\left[\sum_{i=0}^{n}(x_i - a)g_i\right] = \sum_{i=1}^{n} x_i D[g_i]$$

   So:

$$\begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix} = \nabla_f = [D[f]]_{\mathcal{E}} = \begin{pmatrix} \frac{\partial x_1 D[g_1]}{\partial x_1} \\ \vdots \\ \frac{\partial x_n D[g_n]}{x_n} \end{pmatrix} = \begin{pmatrix} g_1 \\ \vdots \\ g_n \end{pmatrix}$$

   So $\frac{\partial f}{\partial x_i} = g_i$ as intended. (Should have done things pointwise for formality but who cares) ■

4. Consider $B \subset \mathbb{R}^n$ the unit ball and suppose $T \colon C^1(B, \mathbb{R}) \to \mathbb{R}$ is linear and $\forall f, g \in C^1(B, \mathbb{R}) \colon T(f \cdot g) = g(0)T(f) + f(0)T(g)$. Then T has the form $T(f) = \frac{\partial f}{\partial u}(0)$.